

1) Inside good conductors such as graphite $\sigma \gg \omega\epsilon$

$$\gamma = j\omega\sqrt{\mu\epsilon} \left(1 + \frac{\sigma}{j\omega\epsilon}\right)^{1/2} \approx \sqrt{j} \sqrt{\omega\mu\sigma} = (1+j) \sqrt{\pi f \mu \sigma}$$

$$\alpha = \beta = \sqrt{\pi f \mu \sigma} \Rightarrow E_x = E_0 \underbrace{e^{-\alpha z}}_{\text{attenuation}} \underbrace{e^{-j\beta z}}_{\text{phase factor}}$$

Skin depth δ denotes the distance wave has traveled when it has been attenuated by a factor of e^{-1} .

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}} \Rightarrow \sigma = \frac{1}{\pi f \mu \delta^2}$$

a) For graphite with $\mu_r \approx 1$, $\mu = 4\pi \times 10^{-7} \text{ H/m}$ at $f = 10^8 \text{ Hz}$

$$\sigma = \frac{1}{4\pi^2 \times 10^{-7} \times 10^8 \times (0.16 \times 10^{-3})^2} = \boxed{0.99 \times 10^5 \text{ (S/m)}}$$

b) 30dB reduction is equivalent to $10^{-\frac{30}{20}} = 0.03162$

$$\text{At } 16 \text{ Hz, } \alpha = \sqrt{4\pi^2 \times 10^{-7} \times 10^9 \times 0.99 \times 10^5} = 1.98 \times 10^4 \text{ 1/m}$$

$$e^{-\alpha z} = 10^{-\frac{30}{20}} \Rightarrow -\alpha z = -\frac{3}{2} \times \ln 10$$

$$z = \frac{1.5 \ln 10}{\alpha} = \frac{1.5 \times 2.3}{1.98 \times 10^4} = 1.75 \times 10^{-4} \text{ m} = \boxed{0.175 \text{ mm}}$$

$$2) \quad \vec{E} = \hat{a}_\theta E_\theta + \hat{a}_\phi E_\phi$$

In last homework Problem 3, we've found that:

$$\vec{E}(\vec{R}) = -\hbar [\hat{a}_k \times \vec{H}(\vec{R})] \quad \text{and} \quad \vec{H}(\vec{R}) = \frac{1}{\hbar} [\vec{a}_k \times \vec{E}] \quad \text{similarly}$$

Choose propagation direction as $\hat{a}_k = \hat{a}_R$ since $\vec{E}$ seems to be a spherical wave. Then

$$\vec{H} = \frac{1}{\hbar} [\hat{a}_R \times \vec{E}] = \frac{1}{\hbar} [(\hat{a}_R \times \hat{a}_\theta) E_\theta + (\hat{a}_R \times \hat{a}_\phi) E_\phi] = \frac{E_\theta}{\hbar} \hat{a}_\phi - \frac{E_\phi}{\hbar} \hat{a}_\theta$$

* Poynting vector $\vec{P}(R, \theta, \phi, t) = \vec{E}(R, \theta, \phi, t) \times \vec{H}(R, \theta, \phi, t)$ denotes the instantaneous power inflicted upon a unit surface,

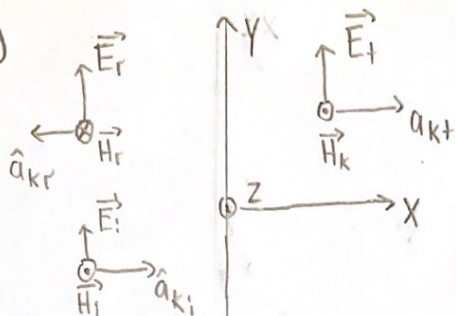
$$\vec{P}_{ave} = \frac{1}{T} \int_0^T \vec{P}(t) dt = \frac{1}{2} \vec{P}_{max} \Rightarrow \text{Ave. power is } \frac{1}{2} \text{ (due to } \cos^2 \text{ integrals) of the peak value or amplitude of } \vec{P}$$

$$\vec{P}_{ave} = \frac{1}{2} \text{Re} \{ \vec{E} \times \vec{H}^* \} \quad \text{in phasor form}$$

$$\vec{E} \times \vec{H}^* = \begin{vmatrix} \hat{a}_R & \hat{a}_\theta & \hat{a}_\phi \\ 0 & E_\theta & E_\phi \\ 0 & -\frac{E_\phi^*}{\hbar} & \frac{E_\theta^*}{\hbar} \end{vmatrix} = \frac{\hat{a}_R}{\hbar} [|E_\theta|^2 + |E_\phi|^2]$$

$$\vec{P}_{ave} = \hat{a}_R \left[\frac{1}{2\hbar} (|E_\theta|^2 + |E_\phi|^2) \right] \Rightarrow \text{Note that ave power is radiating outward radially as expected.}$$

3)



$$\epsilon_r = 1$$

$$\mu_r = 1$$

$$h_1 = 120\pi$$

$$\beta_1 = \frac{\omega}{c} = \frac{10^8}{3 \times 10^8} = \frac{1}{3} \text{ rad/s}$$

$$\epsilon_r = 2$$

$$\mu_r = 8$$

$$z=0$$

$$\sigma = 0$$

$$h_2 = 240\pi = 2h_1$$

$$\beta_2 = \frac{\omega}{c} \sqrt{\mu_r \epsilon_r}$$

$$= 4\beta_1 = \frac{4}{3} \text{ rad/s}$$

$$\begin{aligned} \vec{E}_i &= \hat{a}_y 50 e^{-j\beta_1 x} \\ \vec{H}_i &= \hat{a}_z \frac{50}{h_1} e^{-j\beta_1 x} \end{aligned}$$

phasors with
sine reference
with $\omega = 10^8$

$$a) \frac{E_{r0}}{E_{i0}} = \Gamma = \frac{2h_1 - h_2}{2h_1 + h_2} = \frac{1}{3}$$

$$\vec{E}_r = \hat{a}_y \frac{1}{3} \times 50 e^{-j\beta_1 x}$$

$$\vec{H}_r = (-\hat{a}_x) \times \frac{1}{h_1} \vec{E}_r$$

$$= -\hat{a}_z \frac{50}{3h_1} e^{-j\beta_1 x}$$

$$\vec{E}_r(x,t) = \hat{a}_y \frac{50}{3} \sin\left(10^8 t + \frac{x}{3}\right) \frac{V}{m}, \quad \vec{H}_r(x,t) = -\hat{a}_z \frac{5}{36\pi} \sin\left(10^8 t + \frac{x}{3}\right) \frac{A}{m}$$

↳ Note the 180° phase change due to reflection

$$b) \Gamma = \frac{h_2 - h_1}{h_2 + h_1} = \frac{1}{3} \text{ : reflection coefficient, } \tau = \frac{2h_2}{h_2 + h_1} = \frac{4}{3} \text{ or } \tau = 1 + \Gamma = \frac{4}{3} \text{ : transmission coefficient}$$

$$c) S = \frac{|E|_{\max}}{|E|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{4/3}{1/3} = 2 \text{ : standing-wave ratio}$$

$$d) E_{t0} = \tau E_{i0} = \frac{4}{3} \times 50 = \frac{200}{3}, \quad \vec{E}_t = \hat{a}_y E_{t0} e^{-j\beta_2 x} = \hat{a}_y \frac{200}{3} e^{-j\frac{4}{3}x}$$

$$\vec{H}_t = \hat{a}_z \times \frac{\vec{E}_t}{h_2} = \hat{a}_z \frac{E_{t0}}{h_2} e^{-j\beta_2 x} = \hat{a}_z \frac{200}{3 \times 240\pi} e^{-j\frac{4}{3}x}$$

$$\vec{E}_t(x,t) = \hat{a}_y \frac{200}{3} \sin\left(10^8 t - \frac{4x}{3}\right) \frac{V}{m}, \quad \vec{H}_t = \hat{a}_z \frac{5}{18\pi} \sin\left(10^8 t - \frac{4x}{3}\right) \frac{A}{m}$$

↳ No phase shift upon transmission

4) In normal incidence:

reflection coefficient : $T = \frac{h_2 - h_1}{h_2 + h_1}$, transmission coefficient : $\tau = \frac{2h_2}{h_2 + h_1}$

and $1 + T = \tau$

For $|T| = |\tau|$ we require $|1 + T| = |T|$

It's apparent that we should have $-T = 1 + T$ and $T < 0$ since otherwise $1 + T = T$ has no solution.

Then from $-T = 1 + T$, $T = -\frac{1}{2}$, $|T| = \frac{1}{2}$

* Total electric field at medium 1 is:

$$\vec{E}_1(z) = \hat{a}_x \left(\underbrace{E_{i0} e^{-j\beta_1 z}}_{\text{incident}} + \underbrace{T E_{i0} e^{+j\beta_1 z}}_{\text{reflected}} \right) = \hat{a}_x E_{i0} e^{-j\beta_1 z} (1 + T e^{j2\beta_1 z})$$

* This will be standing wave with $|E|_{\max} = 1 + |T|$
and $|E|_{\min} = 1 - |T|$

Then standing-wave ratio is $S = \frac{|E|_{\max}}{|E|_{\min}} = \frac{1 + |T|}{1 - |T|} = \frac{3/2}{1/2} = \boxed{3}$

Or in decibels: $S = 20 \log_{10} 3 = \boxed{9.54 \text{ dB}}$